Student Absence Tracker

Journal of how the app was built and developed

23	67	3811	33
development phases	published changes	lines in the main page	files in the project

From September 30, 2026 to October 19, 2026

This file last updated: October 23, 2026

About the app

An Arabic (right-to-left) web app for recording student absences period by period on a weekly sheet, shared live by the school's offices, with statistics, notifications to parents, a make-up list and a French version.

It was built over successive conversations with the person in charge of the app: they ask for an addition or a change, it is implemented, tested and published to both sites, then the result is reviewed and a new improvement is requested. This file tells those phases in chronological order.

Technical architecture

Part	Description
Front end	A single HTML file with plain JavaScript (no framework), an fr.js file for instant French translation, and sw.js so the page works offline
Server	Vercel functions in the api folder (accounts, saving the sheet, statistics history, school-year archive)
Storage	Upstash Redis, with a KEY_PREFIX key prefix that isolates each school's data
Deployment	Two Vercel projects from the same repository (Al-Mothabara Private Institution, and Al-Mothabara Private Primary School), deployed automatically on every push to main
Testing	Dozens of automated Playwright browser tests (recording absences, printing, permissions, weeks, offline work, statistics, import…) run before every release

Timeline

Date	Phase
2026-09-30	1. Foundation: the shared absence sheet
2026-10-01	2. Accounts and permissions
2026-10-02	3. Several schools, first statistics, a logo per school
2026-10-03	4. Deeper statistics and confirmation before recording
2026-10-04	5. Semi-automatic notifications to parents
2026-10-05	6. Recording and correction safeguards
2026-10-06	7. Number of periods per school timetable
2026-10-07	8. The big day: logo, approvals, French, colours
2026-10-08	9. Make-up list and hardening “Add a student”
2026-10-09	10. Notifications restricted to the administrator
2026-10-10	11. Working offline and the week buttons
2026-10-13	12. Timing per class, header order and the school year
2026-10-14	13. Importing the teaching organisation from Excel
2026-10-15	14. Absence sheet: pinned columns, bottom rows, the bar and dragging the table
2026-10-17	15. Justification inside the absence cell and its breakdown in the statistics
2026-10-17	16. Mid-term holiday week: a locked sheet
2026-10-17	17. “Teacher” account records absences at roll call
2026-10-18	18. Importing a list without losing absences, and backups of the sheet
2026-10-18	19. Changing the “from” date shows that week's data at once
2026-10-18	20. Training pack: demo site and fictional Excel files
2026-10-18	21. “Nobody” for the teacher, and teachers working without internet
2026-10-19	22. Removal of the icon home screen
2026-10-23	23. Inviting teachers by e-mail: a ready account with one click on the link

Development phases in detail

1 Foundation: the shared absence sheet

2026-09-30

Request: A weekly sheet where every office records student absences and all other offices see them instantly, without a computer acting as a server.

- Weekly table: five days × morning and afternoon periods, a tab per class, and a student list with the parent's phone number.
- Columns for subject and teacher, and totals of absence hours (per period, per day, per week).
- Parent notification through a ready-made WhatsApp message, and printing of the class sheet with the school header and logo.
- Import of a student list from an Excel file (choosing the names column and the header row from a preview).
- Shared storage on Vercel and Upstash Redis with periodic synchronisation between offices, protected by a single password.

Technical notes: First commit: index.html (about 940 lines), api/state.js and package.json.

2 Accounts and permissions

2026-10-01

Request: A single administrator who edits, and view-only accounts that browse without changing anything, instead of one shared password.

- Login with email and password; the administrator is the email set in the server settings and cannot be registered through the public form.
- View-only accounts are read-only: writing is refused by the server itself (403) however the buttons are shown in the browser.

Technical notes: Signed sessions (HMAC-SHA256) valid for 30 days, and passwords stored as salted scrypt hashes. New files: api/auth.js, api/_lib.js and api/_seed.js.

3 Several schools, first statistics, a logo per school

2026-10-02

Request: Use the same code for more than one school with separate data, and see absence figures instead of just the table.

- The KEY_PREFIX, INSTITUTION_NAME and SEED_BLANK variables: each Vercel project gets its own data, accounts and name.
- Statistics page: total absence hours, number of absent students and their rate by day, week and month, per class or level, with printing.
- Weekly archive: on every save a summary of the week is stored on the server (api/history.js), so past weeks stay available and the sheet itself does not grow.
- A logo specific to each school (LOGO_FILE) with a layout for wide logos.

Technical notes: api/_stats.js summarises the week and derives the level from the class name (1AIPC-2 → 1AIPC).

4 Deeper statistics and confirmation before recording

2026-10-03

Request: Richer reports for the administration, and protecting the sheet from a wrong click.

- A per-student view, class and level filters, and charts that are included when printing.
- Export of charts as PNG images and of the table to Excel.
- Every recording or deletion of an absence asks for confirmation that names the student, the class and the period.

Technical notes: About 290 lines for the statistics, and a change to how clicking a cell behaves.

5 Semi-automatic notifications to parents

2026-10-04

Request: Inform parents of absences without anyone counting hours by hand.

- One hour of absence → a notice to the parent; three hours → a summons to come to school.
- The count is cumulative over the month (archived weeks plus the current sheet), and what was already sent is remembered so it is not repeated.
- The message is prepared as ready text in WhatsApp and the administrator only has to press “Send”.

Technical notes: Fully automatic sending needs a paid service provider, so WhatsApp with one click by the administrator was chosen.

6 Recording and correction safeguards

2026-10-05

Request: Prevent common mistakes and correct what was recorded by error.

- An absence cannot be recorded for a period whose time has not come yet, and upcoming cells look different.
- A button to go back to the previous week to correct a mistaken absence; on save, that week's summary in the statistics is updated automatically.
- The days and periods rows stay visible while scrolling the student list, and are repeated at the bottom of the table.

Technical notes: The finished week is kept in full (absences, subjects and teachers) in state.other so it comes back exactly as it was.

7 Number of periods per school timetable

2026-10-06

Request: Fit the table to schools that differ in how many periods they have.

- Add or remove a period in the morning and in the afternoon; subjects, teachers, absences and period times are realigned so nothing slips into a wrong period.
- Archived weeks keep their own layout, so their figures stay correct.

Technical notes: state.pc = [morning periods, afternoon periods], stored with every weekly summary.

8 The big day: logo, approvals, French, colours

2026-10-07

Request: Let the school manage itself and extend use to French-speaking users.

- The administrator changes the school logo from the app; it is stored on the server and shown on every device.
- New view-only accounts wait for the administrator's approval (the “Approve a new account” button).
- The French version: language switching within the same page and left-to-right direction, covering WhatsApp messages, printing, charts and Excel (the fr.js file).
- The interface is named “Student Absence Tracker”, and so is the browser tab.
- A thick separator between morning and afternoon and after each day in the table and in print, and a colour layer (day colours, coloured buttons and the week bar).
- Buttons rearranged: the tabs row below the buttons, and list import next to “Add a student”.

Technical notes: fr.js is a translator that runs at runtime: a dictionary for fixed phrases and patterns for sentences with numbers, plus a page observer that translates whatever is added to it.

9 Make-up list and hardening “Add a student”

2026-10-08

Request: Let the support team schedule make-up sessions, and deal with a complaint about the “Add a student” button.

- “Make-up list”: absent students and the subjects they missed (from the weekly archive), with Excel export.
- Telling parents the date, time and place of the make-up session through WhatsApp, which also works for view-only accounts.
- “Add a student” button: the cursor stays in the name field, a warning for an empty name, readiness for the next name, and preventing local storage or an old poll from cancelling the save.

Technical notes: The fault could not be reproduced on our side, so a preventive hardening was added, then the user confirmed that everything works.

10 Notifications restricted to the administrator

2026-10-09

Request: View-only accounts must not send absence notices or summonses.

- The automatic-notice and summons panels and their send buttons are shown to the administrator only.
- A clearer frame around the new student's name field.

Technical notes: Make-up session notices remain available to view-only accounts, as requested earlier.

11 Working offline and the week buttons

2026-10-10

Request: Keep recording absences when the network drops, and simplify moving between weeks.

- Unsent edits are kept on the device and sent automatically when the connection returns; the page opens without internet, with a bar showing the status and a warning window if another administrator edited in the meantime.
- The week buttons sit under the two date fields and are named “Previous” and “Next”, and the “Current week” button comes before the date on its row.

Technical notes: A copy of the sheet in localStorage with “unsent” and “copy base” markers; sw.js caches only the page shell and never caches /api. A test showed that edits were being erased when the connection returned, so the design was corrected.

12 Timing per class, header order and the school year

2026-10-13

Request: Finer detail per class, then a whole school year that can be archived.

- Period times per class, and a class without settings follows the default times.
- Header order: “Logo” before “Current week”, and “Log out” at the end of the first row.
- The school year appears at the top of the page (inferred from the date or edited by the administrator).
- “Archive the season” moves all weeks' absences into the season archive and starts the statistics from zero in the next season, keeping the lists and subjects.
- A “Season” selector in the statistics to browse archived seasons (read, print and export), and deleting them is for the administrator.
- The “Archive the season” button sits under the table next to “Add a student”.

Technical notes: api/season.js: each week is stored in its own field in Redis, and archiving a season twice is prevented. Two corrections for our misunderstanding of two requests: the “Change season” button was meant to name the season, not to archive it.

13 Importing the teaching organisation from Excel

2026-10-14

Request: Stop typing the subject and teacher name by hand for each period of each class.

- The “Import teaching organisation” button reads the “teaching organisation” file: teacher blocks (code → teacher + subject), the classes table and the “timetable for class” sheets.
- A window decides, for each class in the file, which class it applies to (match by name, create a new class, or ignore).
- Spelling mistakes are corrected from the class table, and names are cleaned of elongation characters and repeated spaces.
- Student lists and absences are not changed.

Technical notes: Tried on the real file: 6 classes, 15 teachers and 28 periods per class, using a substitute Excel reader in the test because downloading the real reader is blocked in the development environment.

14 Absence sheet: pinned columns, bottom rows, the bar and dragging the table

2026-10-15

Request: The phone column disappeared when moving between days of the week, unlike the names column. Then a request to move the table left and right by long-click and drag. Then the “Add a student”, “Import list from Excel” and “Archive the season” buttons were to stay visible while the table moves, with the archive button at the far left of the row. Then the “absence count per period” and “days of the week and periods” rows below the list were to stay visible too. Finally the “subjects” and “teachers” row was to be pinned as well.

- The “Parent's phone” column stays visible next to the names column while the sheet scrolls horizontally between days.
- The two column headings (“Full name” and “Parent's phone”) stay visible as well.
- Moving the table with the mouse: a long left-click then drag left or right (and up/down), with the cursor turning into a grabbing hand.
- Dragging never records an absence by mistake; a normal click opens the confirmation window as usual, and typing in the phone fields and the buttons is unaffected.
- The add bar (new student's name + the three buttons) is now fixed at the bottom of the table and stays visible however the table is moved sideways or the list scrolls down.
- The “Archive the season” button is at the far left of the row, and the bar wraps onto two lines on narrow screens.
- The “absence count per period” row and the “days of the week and periods” rows below the list are now pinned directly above the add bar, and stay visible while moving around the table or the names and phones list.
- The subjects and teachers row is now pinned under the days and periods rows at the top of the table while going down the list.

Technical notes: position:sticky for the phone column with an offset equal to the names column width, measured after every table render (the --namew variable), so it is correct with long names. Dragging: mousedown, mousemove and mouseup events with a 5-pixel threshold, and the click following a drag is cancelled (touch is unaffected since it already scrolls). The bar sits inside a wide wrapper with position:sticky (bottom and right) whose width equals the visible area (the --hostw variable). The heights of the pinned rows are measured after every render (the --b0, --b1 and --b2 variables) so they stack from the bottom. The three top headings stack by measuring their heights (--hr1 and --hr12), and the pinned label cells cover only the name and phone columns so they do not hide periods on narrow screens.

15 Justification inside the absence cell and its breakdown in the statistics

2026-10-17

Request: A “justified or unjustified” choice is more useful if added in the absence-recording cell itself, next to “A”. (It was first tried as a per-student column before the phone, then the column was removed and this solution adopted.) Then a breakdown of justified and unjustified absences in the statistics was requested.

- Clicking an empty cell makes the confirmation window ask: “Record unjustified absence” or “Record justified absence”.
- An unjustified absence shows a red “A” as before, a justified one shows “A” with a small “J” in green, and a tooltip explains the state on mouse hover.
- Clicking a recorded absence lets you delete it or switch it between justified and unjustified without notifying the parent again.
- A justified absence counts in the counters and statistics like any other, and is saved in the weekly history, so it comes back as it was when returning to the previous week.
- In print it appears as a small “J” above the “A”, and in the French version as “absence justifiée / non justifiée”.
- Main cards: under “Absence hours (today)” and “Total absence this week” two badges: how many hours are unjustified and how many justified; and under “Absence by subject / teacher” the phrase “of which n justified”.
- Statistics window (day / week / month, by class / level / student): two new columns “of which justified” and “of which unjustified” after “Absence hours”, in the table, in print and in the Excel export.

Technical notes: The cell value became true (unjustified) or 'j' (justified), so every calculation built on truthiness stays correct. In the server's weekly summary (api/_stats.js) there is a digit per period: 0 present, 1 absent, 2 justified absence. The server's weekly summary carries a colsJ array (justified per period) next to cols; weeks saved before the feature are counted as entirely unjustified.

16 Mid-term holiday week: a locked sheet

2026-10-17

Request: The week of 19 to 23 October 2026 is a mid-term holiday: it stays locked, nothing can be recorded, and a prominent “Mid-term holiday” text appears across the days of the week.

- When this week is shown, the absence cells are hidden and replaced by large hatched blocks reading “Mid-term holiday” above each half-day (morning / afternoon); the subject and teacher cells are locked and clicking opens no window.
- The locked week does not enter the statistics (its summary is not saved and it does not count as a recorded day), and moving on from it with “Next” does not replace the real “previous week”.
- Printing shows the same wording instead of the cells; the French version reads “Vacances intermédiaires”.
- Incidental fix: printing a sheet containing a justified absence used to break, and was corrected.

Technical notes: The list of weeks is in state.breaks (from / to / label), with a default value in the interface and in api/_stats.js (summarizeState returns empty for the holiday week). There is no screen yet to edit the list, so adding another holiday is done by editing the data.

17 “Teacher” account records absences at roll call

2026-10-17

Request: Trial of giving the teacher permission to record absences at roll call at the start of the period, without being able to edit the rest of the sheet.

- A third role next to administrator and view-only: the administrator turns any approved account into a “teacher” from the accounts window, with an optional teacher name as it appears in the table, and withdraws the permission with one button.
- The teacher sees the sheet read-only, but the absence cells become clickable in today's periods whose time has started: one tap records the absence with no questions, and a tap on a recorded absence asks to confirm deletion. If a name is saved, the teacher is limited to their own periods.
- The teacher interface is simplified for the phone: only the season, the week date, the class tabs bar, the class table with its students' names, an “Add a student” button, the language switch and logout; statistics, printing, make-up, import, approval, parent notifications, phone columns and deletion are hidden; the language and logout buttons are on one line on a phone screen.
- The teacher adds a student by name only through the server (which refuses duplicates and empty names), and cannot delete or edit; parents' phone numbers are never sent to them by the server.
- A justified absence is the administrator's decision alone, so the teacher neither creates nor removes it, and sees no edit buttons or parent notifications.
- The server enforces all of this (today's date in Casablanca time, the period has started, not a holiday week, matching the student's name and place), and saving the whole sheet stays for the administrator only.
- Two simultaneous roll calls by two teachers do not lose one another, because the server locks reading and writing the sheet; this was tested with six parallel requests.
- The French version of all new texts and messages.
- Both cases verified: a teacher account with no saved name records in any period that has started today, and an account with a saved name records only in their own periods. Name matching now tolerates a different order, titles, spellings of alef, ya and ta marbuta, and accented Latin letters, with the same rule on the server and the page.
- When a teacher taps a locked cell a message explains why (weekend, sheet on a past week, another day, period not started, another teacher's period with both names, justified absence), in Arabic and French, instead of leaving the cell silent.

Technical notes: A new endpoint api/mark.js edits a single cell inside the stored sheet (read then write under a Redis SET NX lock). The role is saved in the account record and overrides what is in the signed token (refreshed within a minute). A known limit: if the administrator saves an old sheet at the very same moment it could overwrite a teacher's roll call, so the server uses the same lock and the

administrator's page resynchronises on its first poll. api/mark.js adds the addstudent action, api/state.js returns parents' phones empty for a teacher account, and hiding in the interface uses the body.tch class.

18 Importing a list without losing absences, and backups of the sheet

2026-10-18

Request: Importing a list that was almost the same as the existing one wiped the week's absences and the phone numbers. Needed: that it never happens again, and a way to recover what gets lost.

- Import now has three buttons: “Update the list keeping absences and phones” (first), “Add”, and “Replace”, which now asks for an explicit confirmation stating how many absences and phones will be erased.
- The update matches names word by word regardless of order and of differences in how alef, ya and ta marbuta are written, and accepts a difference of one or two letters when it is not ambiguous; whoever is in both the file and the list keeps their absences and phone, and the administrator is asked about those not in the file (delete them or keep them). “Add” no longer duplicates an existing name.
- Automatic backups on the server before any save that loses absences, phones or students in one go, and at most every 12 hours, keeping the last 15 copies.
- A “Backups” button for the administrator: list the copies, save one immediately, and restore partially (absences and phones for the students of the current list, matched by name) or fully; every restore first saves a copy so it can be undone.
- Incidental fix: confirmation messages now appear above the import window instead of hiding beneath it.
- The French version of all new texts.

Technical notes: api/_backup.js (snapshot on loss or periodically, the index, restore by matching) and api/backups.js under the same lock used by api/state.js. The snapshot is taken in POST /api/state after reading the stored state. What was lost before this addition has no copy in the app; what may exist in Upstash backups or in the weekly statistics archive (for previous weeks only) remains.

19 Changing the “from” date shows that week's data at once

2026-10-18

Request: When the week's start date is changed in the “from” field and Enter is pressed, the table's data changes to match the chosen week instead of only relabelling it.

- If the new date is within the same week (the same Monday) only the label changes and the data stays.
- If it is in another week, the current week is saved first and then the chosen week is shown: in full (absences, subjects and teachers) if it is the other week kept in the sheet, otherwise rebuilt from the statistics history (absent students, subjects and teachers). The week that was displayed stays saved, so typing its date again restores it in full.
- The mid-term holiday week opens directly, locked.
- For a week with no data the administrator is asked to choose: show an empty week at this date, or change the date only while keeping the current data, or cancel (the old date returns). An empty or invalid date restores the previous one.
- An empty future week is not added to the statistics archive (so the statistics do not fill up with empty weeks).
- The French version of the new texts.

Technical notes: jumpToWeek and archivedWeekAt in index.html (using state.other, snapshotWeek, applyWeek and ntLoad), and a guard in archiveWeek in api/_stats.js that skips a week which has not started yet and has no absences. A new Playwright test covers: a previous week, going back, the same week, a week with no data (3 buttons), the holiday week, an empty date, and the server state.

20 Training pack: demo site and fictional Excel files

2026-10-18

Request: Make a training course on the app possible with a demo site and a completely fictional school, without touching any real school’s data or using real names.

- New run mode: SEED_DEMO in the Vercel settings. Value 1 starts with an empty timetable that trainees fill in by importing; value 2 starts with a ready timetable. On the first visit the “Al-Ofoq (demo)” school is created: 6 classes, 121 students with invented names and invented phones (starting with 06000), an empty current week and four past weeks of invented absences on the statistics page.
- The setting only acts when the database is empty, so it cannot change the data of an existing project; the demo site is kept apart from the others by its own KEY_PREFIX.
- Three fictional Excel files (docs/training): student lists (one sheet per class), an updated version (one student leaves each class, newcomers join, alphabetical order) for the “update while keeping absences and phones” exercise, and a fictional “educational organisation” file (class grid with lesson codes, timetable with subjects, blocks for the fifteen teachers) with no teacher in two classes at once.
- The two generators (export.js and build-xlsx.py) read the same data as the server, so the files and the demo site always agree.
- README guide in docs/training: creating the training Vercel project, resetting after each course, answer key for the exercises and a table of the fictional teachers.

Technical notes: File api/_demo.js (deterministic data, fixed-seed random generator, timetable built by random search with no teacher clashes) and a modified GET in api/state.js (set with NX, then the weeks written to history). Playwright tests with a stub Excel reader: seeding, importing the educational organisation (result identical to the SEED_DEMO=2 timetable, lesson by lesson) and importing the updated list (absences and phones kept even though positions changed).

21 “Nobody” for the teacher, and teachers working without internet

2026-10-18

Request: Let the administration check that the teacher did the roll call even when nobody is absent, and keep absence recording working when the teacher’s internet goes down.

- A “Nobody” (لا أحد) button under each period for the teacher: pressed when the roll call is done and nobody is absent, it records the teacher’s name and the time on the sheet. One tap plus a confirmation cancels it; it cancels itself if an absence is later recorded in that period, and it is refused if an absence is already there. The rules are those of recording an absence: today’s period, already started, and the teacher’s own.
- A second “✅ Nobody” button in the add-student bar for the teacher: it acts on their period in progress in the class (the latest started period of today that is theirs) and asks for a confirmation naming the class, period and subject; for the administrator, the “✅ Roll-call follow-up” button after “Import a list from Excel”.
- Follow-up for the administration: under each period’s absence count, a ✓ (roll call done with “Nobody”) or a red “!” (no absence and no “Nobody” 20 minutes after the period began); the “✅ Roll-call follow-up” button lists the periods without a recorded roll call (class, period, subject, teacher) and the “Nobody” declarations with their time, for today or the whole week. It covers the current week and periods that have a subject or a teacher.
- Teachers working without internet: the app opens on the device where the teacher has already signed in (account, name and last sheet are kept) and they record absences and “Nobody” as usual; each tap shows at once with an orange dashed frame (“waiting”), the status bar shows how many are unsent, and everything is then sent automatically, in order, when the network returns.
- Deferred records carry the time of the tap: the server checks “today” and “period started” against that time rather than the arrival time (and refuses a future time or one older than 18 hours). Whatever the server refuses on arrival (an absence justified in the meantime, a new week, a student removed from the list) is reported to the teacher with the student, the period and the reason, and the screen returns to the server’s copy. If the session expires, the records stay and are sent after signing in again with the same account; the “Log out” button warns when some remain.
- Complete French interface for these additions, and the README updated (the “teacher account” and “working without internet” sections used to say the teacher needs internet).

Technical notes: In api/mark.js: a new action 'called' storing columns[i].called={at,by}, a shared teacherRules function for the teacher’s rules, an optional at field with tapTime (18-hour ceiling, future refused), and the marker removed when an absence is recorded. In index.html: a local queue ghiyab_tq (operations carry absolute values so a resend is harmless; the last tap on a cell replaces earlier ones), a

tqOverlay layer re-applied over every server copy so automatic sync never erases what is not yet sent, tqFlush sending in order and stopping on a dropout or 5xx, dropping refusals for good (409 stale retried three times), and the teacher’s name kept in USER_KEY so the app opens offline with their rights. Tests: 28 server checks (rules and times) and a browser test with simulated dropout, page reload with no network, automatic sending, refusal on arrival and the logout warning, plus a French test checking that no Arabic text is left on screen.

22 Removal of the icon home screen

2026-10-19

Request: Withdraw a trial that did not help users.

- A home screen of icon tiles (17 tiles for the administrator) was tried and then removed at the administrator’s request: most tiles opened the same interface as before and added nothing. The interface is back as it was: the app opens straight on the sheet and the statistics, catch-up and other buttons are where they were, with no effect on the server or the data.

Technical notes: Undone with git revert of the home-screen commit (index.html, fr.js, README).

23 Inviting teachers by e-mail: a ready account with one click on the link

2026-10-23

Request: So that teachers no longer wait for an approval: they receive a link by e-mail and their account becomes a teacher account as soon as they open it, while the “Approve a new account” button stays for unknown people who registered themselves.

- In the “Approve a new account” window, an “Invite teachers” box: one line per teacher (the e-mail, then the name as in the timetable, optional) and a “Send the invitations” button that handles several lines at once and skips duplicates. Below it, the “Invitations waiting” list with “Resend” and “Cancel”.
- The link is random, valid for 7 days and works once. The teacher opens it and the “Activate the teacher account” page shows the locked e-mail; they choose a password (at least 8 characters) and go straight in with an approved teacher account (tied to their name if the administration entered it). If the link is opened again or has expired, the ordinary login page appears with the reason.
- The e-mail goes out through any SMTP account (for example Gmail with an app password), set with Vercel variables, without any external library. If nothing is configured or sending fails, the window shows the link with “Copy the link” and “WhatsApp” so the administrator sends it themselves: the feature always works, e-mail is only a convenience.
- Security: only the SHA-256 of the link is stored; activation can be used once even with two simultaneous requests; an account request filed by someone who does not own the address is removed at invitation time (or its password is replaced by the link holder's); the link never changes the password of an already approved account; attempts are limited per IP address; the SMTP password never travels over an unencrypted connection. An already approved account is simply promoted to teacher by the invitation, with no link.
- Complete French translation (window, activation page, messages) and a fix in fr.js: text typed into a text area stays as typed but its hint (placeholder) is now translated. README updated with the “Inviting teachers by e-mail” section and the Gmail setup.

Technical notes: New file api/_mail.js (a small SMTP client on net and tls: direct TLS on 465 or STARTTLS on 587, AUTH PLAIN/LOGIN, base64 multipart MIME message with UTF-8 headers, 9 s timeout). In api/_lib.js, the invitation helpers (loadInvite, hashToken, keys <KEY_PREFIX>:inv:<hash> with a 7-day ex and <KEY_PREFIX>:invites as the administrator's index). In api/users.js, the invite and uninvite actions and the invites and mail fields in GET ?all=1; in api/auth.js, the accept action (the token is consumed with DEL and only the request that really deleted it goes on) and GET ?invite=. In index.html, the invite mode of the login screen and invParse/invSend/invShow. Tests: 40 server checks (including simultaneous use and address squatting), 14 for the mail client against a fake SMTP server in its three modes, and interface checks in Arabic and French at phone width.

Working method

1. Every change is built first in a local test copy, and its tests are run together with the old tests before release.
2. Permissions never rely on hiding buttons alone: every write is checked by the server.
3. The archive does not enlarge the sheet that is polled every second and a half: separate summaries are kept on the server.
4. The French translation covers every new text (dictionary + sentence patterns) and is checked by a test.
5. When a request is ambiguous (such as where a button goes), the closest understanding is implemented, then corrected according to the user's feedback, and the reading that was understood is stated.

Still open

- Delete the old Vercel domains by hand (up to the user).
- Try “Archive the season” for the first time at the real end of the season, because clearing absences from the table cannot be undone.
- Try the teaching-organisation import on the live site with the real Excel reader.
- Try the teacher account in a real lesson: is a single tap without confirmation appropriate, and must the account be linked to the teacher's name in every account? Also try the “Nobody” button (is the 20-minute delay before the “!” right?) and working without internet in a room with no coverage.
- Try restoring from “Backups” on real data, and enable the daily backup in Upstash if the plan allows it.
- Set up e-mail sending (SMTP) on Vercel and try a real invitation to a teacher, checking that the message lands in the inbox and not in spam.

Appendix: change log

Hash	Date	Change
89e7d1d	2026-09-30	Add files via upload
7a52e25	2026-10-01	Add authentication: admin edits, registered viewers read-only
3bfa1c2	2026-10-02	Support several institutions: KEY_PREFIX, INSTITUTION_NAME, SEED_BLANK env vars
7de4c57	2026-10-02	Add absence statistics (daily/weekly/monthly, by class and level) with printing
ba81098	2026-10-02	Per-institution logo (LOGO_FILE) with wide-logo layout; add Al Marjaya logo
9f76aab	2026-10-03	Statistics: per-student view, class/level filters, charts, PNG/Excel export, charts in print
a9cea1d	2026-10-03	Ask for confirmation before recording or removing an absence
2639295	2026-10-04	Automatic parent notices: 1 hour -> notice, 3 hours -> summons (monthly total, WhatsApp)
41773dc	2026-10-05	Do not allow recording an absence in a period that has not started yet
e466bcc	2026-10-05	Button to go back to the previous week and fix a mistaken absence
a9451ce	2026-10-05	Keep the days/periods rows visible while scrolling the student list and repeat them at the bottom
6feaf53	2026-10-06	Add/delete periods per half-day (morning/evening) to fit the school's timetable
223c25d	2026-10-07	Let the admin change the institution logo (stored on the server, synced to all devices)
f139d21	2026-10-07	Require admin approval for viewer accounts
5088d3e	2026-10-07	Show the class tabs row below the function buttons row
e83d7cf	2026-10-07	Toolbar: import button under the class tabs, logo button first, periods before times
f8b641b	2026-10-07	Import button beside 'add student'; account requests button under statistics
390978d	2026-10-07	Rename the account requests button to 'الموافقة على حساب جديد'
32cd786	2026-10-07	French interface: language switch (fr.js), LTR layout, French WhatsApp messages, prints, charts and Excel export
2e629a2	2026-10-07	Rename the Arabic interface title to 'تتبع غياب التلاميذ'; fix French hours pattern
96de0df	2026-10-07	Browser tab title: تتبع غياب التلاميذ
0c96967	2026-10-07	Thick vertical separator between morning and evening periods (table and printed sheet)
b597905	2026-10-07	Thick vertical line after the last period of every day
8fd698c	2026-10-07	README: etablissement-secondaire address
d8d222f	2026-10-07	Colour layer proposal: weekday hues, coloured action buttons, week ribbon (preview branch)
46b3e9b	2026-10-08	Catch-up list (قائمة التعويض): absent students and the subjects they missed, for the support team
e27f643	2026-10-08	Add-student button: keep focus in the name box while pressing, feedback on empty name, ready for next name; never let browser storage or a stale poll block/undo a save
c26ec7f	2026-10-08	Catch-up list: tell the parents the date, time and place of the catch-up session via WhatsApp (works for read-only accounts)

2dd4224	2026-10-09	Absence notices and summons to parents: administrator only (read-only accounts no longer see the panels or send buttons)
d737ab4	2026-10-09	Add-student name box: clearer border and focus ring
a776f6f	2026-10-10	Work without internet: unsent edits are kept on the device and sent automatically on reconnect; page opens offline (service worker); conflict prompt; offline status banner
81edcd4	2026-10-10	Week buttons («الأسبوع الماضي» and «أسبوع جديد») directly under the week date fields
acf865f	2026-10-10	Rename week buttons to السابق / التالي and add الأسبوع الجاري button beside the dates
a1a9e21	2026-10-10	Move logo button above the current-week button; wrap week controls on narrow screens
322fe74	2026-10-13	Per-section period start times; logo button on the same line before the current-week button
a65c26f	2026-10-13	Header: logout button ends the first row (after the user name), other buttons start a new row, as in the French layout
18fb65e	2026-10-13	School year: show the season and archive the annual absence (api/season.js, archive button, season picker in statistics)
7e4bd4c	2026-10-13	Toolbar: season archive button goes last in the row
77a8e2a	2026-10-13	Season label at the end of the subtitle line; archive button back between periods and times
af9d148	2026-10-13	Move the season archive button under the table, next to add-student and import
9206a40	2026-10-14	Import the teaching organisation (subject + teacher per section) from the Excel timetable file
d6414b5	2026-10-14	Add development journal PDF (auto-generated from docs/journal.json) and update rule
e42b138	2026-10-15	Keep the phone column visible beside the name column when scrolling between days
d522640	2026-10-15	Drag the attendance sheet sideways with the mouse (click and drag)
10fef07	2026-10-15	Keep the add/import/archive bar pinned while moving the sheet; archive button at the far left
732da02	2026-10-15	Pin the absence-count row and the days/periods rows above the add bar
af91154	2026-10-15	Keep the subjects/teachers row pinned under the days and periods rows
16309fb	2026-10-15	Development journal PDF: use Latin digits only
998ac5d	2026-10-17	Add a justified / unjustified toggle in front of each parent's phone number
94bf6be	2026-10-17	Remove the per-student justified/unjustified column (replaced by a per-absence marker)
f5bb051	2026-10-17	Justified / unjustified absence inside the attendance cell (غ red, م+غ green)
9d28d47	2026-10-17	Statistics: detail justified / unjustified absences (cards, tables, exports)
7c4adc2	2026-10-17	Lock the inter-term holiday week (19-23 Oct 2026): show 'عطلة بينية', keep it out of the statistics; fix printing a justified absence
42b76bd	2026-10-17	Teacher accounts: record absences at roll call (server-enforced via /api/mark: today, started period, own name, no justified); admin grants it from the accounts dialog
3ddf135	2026-10-17	Teacher view: simplified mobile layout (season, week dates, class tabs, sheet, add student, language, logout), add-student for teachers via /api/mark, parents' phones hidden from teacher state
452c155	2026-10-17	Teacher phone header: language and logout buttons on one line

7eb5f8e	2026-10-17	Teacher name matching: tolerant of word order, titles, alef/ya/ta-marbuta spellings and Latin accents (same rule on server and page)
17efc57	2026-10-17	Docs: teacher name matching rules, journal
3755430	2026-10-17	Teacher: explain why a locked cell cannot be recorded (weekend, stale week, other day, not started, not your period, justified); server message names both teacher names
2537074	2026-10-18	Import keeps absences and phones for matching names (sync/append/replace with confirmation); automatic server snapshots before data-losing saves, with partial/full restore dialog
e9fb3e2	2026-10-18	Week date: typing a new 'from' date + Enter shows that week's data (kept/archived/blank/holiday); empty future week not archived; docs, journal PDF
6e5584c	2026-10-18	Development journal PDF in French and English (builder renders ar/fr/en from journal.json + translations); rules updated
8608c19	2026-10-18	Training pack: SEED_DEMO demo site, fictional student lists and educational-organisation Excel files
65a2472	2026-10-18	Teacher roll call «لا أحد» with admin follow-up, and teacher work without internet (local queue, auto-send, tap-time rules)
50c45c7	2026-10-18	button in the add-student bar for teachers (current period, with confirmation); «لا أحد» follow-up button after the import button for the admin
9b9d905	2026-10-19	Home screen with colourful icon tiles for the administrator and read-only accounts (teachers keep going straight to the sheet)
ad45f16	2026-10-19	Revert «Home screen with colourful icon tiles» (requested by the administrator); journal phase 22 records the withdrawal

Commit messages are kept in English as written. This file is generated automatically by node docs/build-journal.js from docs/journal.json, its translations and the git log.